What is it?

While it all starts with a new chip and the software in it, embedded design does not end there. Rather the production of a chip is just the beginning of innovations. The same chip can be used on different circuit-boards to produce everything from a piece of medical equipment, a highly sophisticated weapon to the very homely set-top box!

To assemble such a fundamental component of our daily lives requires fascinating and complex tools, namely, electronic design automation (EDA) tools. If slicing, analysing and understanding technology gives you a kick, then breaking down an EDA tool or rather putting one together can be a love story by itself.

Key drivers of growth

Embedded design is all about designing new software and putting together the hardware to execute it all. The field is wide and grew by 30 per cent on a year-on-year basis in 2006-07 — from $4.6 billion (Rs. 240 billion) to $6 billion (Rs. 184 billion). Embedded software contributed $4.9 billion (Rs. 196 billion) and employed 106,000 engineers, accounting for 82 per cent of the total embedded design services workforce, according to an industry report.

Why is it needed?

Recession has taken its toll on this industry. As the sales of home appliances dropped, so did the demand for the chips inside them. The industry believes that a recovery looks likely only by mid-2010. While there will be a fresh demand for already-existing electronic goods, there is a next generation of chips that is already on the table. Chips of the future will help us improve the efficiency of generating electricity more effectively from sunlight, water currents and any other renewable energy source. Medical equipment to mobile phone gaming, all of these will improve. And chips that might just extend the science of motion-gaming to a whole new array of products.

Earlier, and to a large extent even today, companies like Intel and Texas Instruments, two of the largest employers in the sector, would get the design for a product from Europe or the US. India would just be the ground for design execution or software development. But as the Indian market grows in size and importance, products are being designed especially for the Indian consumer.

One of the more common examples being the set-top box: What started with one state, i.e. Tamil Nadu's experiment with the cable network, has now spread out to be an entire market in itself. Tata was the first entrant followed by Reliance, Airtel and Sun Direct. At the time of its launch, India did not have the capacity either to design or manufacture the integrated circuits (ICs) required in the set top boxes (STB). Import was the only route. But as more market players entered the fray, the demand for a home-grown product increased. Government also did its bit by offering incentives to chip manufacturers. The result — the STM 5107 from ST Microelectronics, a geode processor from National Semiconductor and Broadcom's system-on-a-chip (SoC) solution — all these chips developed right here at the Indian Design Centres and already in use by various set-top box providers. Market researcher Instat says that by 2012, India's pay-TV market is expected to reach 90 million subscribers. It estimates upwards of $1.0 billion worth of market for STB ICs by the year 2012.

Any of the above-mentioned chip manufacturers provide their buyers